

Day 19 - Lecture 19.3: Vertical Reference Lines (`plt.axvline`)

1. Overview & Purpose

In data analysis, histograms show the distribution, but stakeholders need **benchmarks and reference thresholds**. For example:

- *What is the pass mark?* (e.g., 33 marks)
- *What is the mean / average score?* (μ)
- *What is the median or 95th percentile?*
- *What are control limits in manufacturing?*

`plt.axvline()` draws an **infinite vertical reference line across the axes** at a specified x-coordinate.

2. Syntax & Key Parameters

```
plt.axvline(  
    x=0,                                # The x-coordinate where the line will be drawn  
    ymin=0, ymax=1,                    # Normalized y-span (0 = bottom, 1 = top)  
    color='red',                        # Line color  
    linestyle='--',                    # Style: '--' (dashed), '-' (solid), ':' (dotted), '  
    linewidth=2,                       # Thickness of line  
    label='Threshold Name'             # Label for legend  
)
```

Companion Methods:

- `plt.axhline(y, ...)` : Draws horizontal reference line across entire width.
 - `plt.axvspan(xmin, xmax, ...)` : Highlights a vertical region (e.g. critical zone).
 - `plt.text(x, y, "label")` or `plt.annotate()` : Places descriptive text directly next to the line.
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3. Mathematical Reference Lines

Given a dataset $X = x_1, x_2, \dots, x_n$:

1. Sample Mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

2. Standard Deviation Bands:

$$\text{Upper: } \bar{x} + \sigma, \quad \text{Lower: } \bar{x} - \sigma$$

Plotting $\bar{x}$ as a vertical dashed line instantly reveals skewness:

- If Mean $>$ Median, distribution is **positively (right) skewed**.
 - If Mean $<$ Median, distribution is **negatively (left) skewed**.
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4. Code Implementation

```
import matplotlib.pyplot as plt
import numpy as np

marks = [
    25, 38, 42, 55, 61, 67, 72, 78, 83, 88,
    91, 95, 47, 53, 59, 64, 69, 74, 79, 85,
    92, 97, 33, 45, 58, 62, 71, 76, 81, 89
]
bins = [0, 30, 70, 90, 100]

plt.figure(figsize=(9, 6))
plt.hist(marks, bins=bins, edgecolor="black", color="lightblue")

# 1. Reference line for Passing Marks (33)
plt.axvline(
    x=33,
    color="crimson",
    linestyle="--",
    linewidth=2.5,
    label="Pass Mark (33)"
)

# 2. Reference line for Mean Score
mean_val = np.mean(marks)
plt.axvline(
    x=mean_val,
    color="darkblue",
    linestyle="-.",
    linewidth=2,
    label=f"Mean Mark ({mean_val:.1f})"
)

plt.xlabel("Marks", fontsize=12)
plt.ylabel("Number of Students", fontsize=12)
plt.title("Distribution of Marks of 30 Students with Threshold Lines", fontsize=12)
plt.legend(loc="upper left", fontsize=11)
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()
```

5. Summary & Best Practices

- Always add `label=` to `axvline` and invoke `plt.legend()` .
- Use contrasting styles (e.g. dashed red) so the threshold stands out clearly against the histogram bins.